

Algorithm infix_to_prefix(A, P, S)

1. Reverse the infix expression
2. Scan this Expression from left to right
3. If the scanned character is an operand then ENQUEUE it to P
4. Else if the scanned character is ')' then PUSH it in to S
5. Else if the scanned character is '(' then
 1. Repeatedly POP elements from S until the popped item is '('
 2. ENQUEUE all these popped items into P except '('
6. Else
 1. POP all the operators from the stack which are greater precedence than that of the scanned operator until S[top] is '(' or S is empty.
 2. ENQUEUE all these popped items into P
 3. Push the scanned operator to S.
7. Repeat steps 3-6 until the entire expression is scanned.
8. Repeatedly POP the remaining elements from S and ENQUEUE it in P
9. Reverse P

PREFIX EVALUATION -ALGORITHM

Algorithm prefix_eval(P)

1. Scan this string from right to left
2. If the scanned character is an operand then PUSH it into S
3. Else
 1. POP two operands(1st popped item is operand 1 and 2nd popped item is operand 2)
 2. Perform the corresponding operation
 3. PUSH result in to S
4. Repeat steps 2-3 until the entire string is scanned.
5. Print S[0]